

## Module 2: Statistical inference (I)

Benjamin Smith

07/08/2026

# Outline

This module we will review

- Basics of probability
- Fundamental concepts in inference

# Probability distributions

- In statistics, we try to draw conclusions about a larger population from a sample of observations.
- We use mathematical models to capture probabilistic behavior of a population.
- This behavior is modeled using probability distributions.

# Density/Distribution functions

## Definition (Cumulative Distribution Function)

$$F_X(x) = P(X \leq x) \quad \forall x \in \mathbb{R}$$



## Density/Distribution functions (cont'd)

### Definition (Probability Mass Function)

For a discrete  $RV$ , the probability mass function (PMF) is:

$$f_X(x) = P(X = x) \quad \forall x \in \mathbb{R}$$

### Definition (Probability Density Function)

For a continuous  $RV$ , the probability density function (PDF) is:

$$f_X(x) = \left. \frac{\partial}{\partial t} F(t) \right|_{t=x}$$

So  $F_X(x) = \int_{-\infty}^x f_X(t) dt \forall x \in \mathbb{R}$ .

Note that  $f_X \geq 0$  for  $\forall x$ , and thus  $F_X$  is an increasing function.

# Expectation and Variance

## Definition (Expectation)

A measure of central tendency (a weighted average of the values of  $X$ )

$$E[X] = \sum_{x \in S} xP(X = x) \text{ for discrete RV taking values from } S$$

$$E[X] = \int_{-\infty}^{\infty} xf_X(x)dx \text{ for continuous RV}$$

## Definition (Variance)

A measure of the spread of a distribution

$$\text{Var}(X) = \sum_{x \in S} (x - E[X])^2 P(X = x) \text{ for discrete RV}$$

$$\text{Var}(X) = \int_{-\infty}^{\infty} (x - E[X])^2 f_X(x)dx \text{ for continuous RV}$$

# Discrete random variable

A discrete random variable has a countable number of possible values.

# Bernoulli and Binomial random variable

- Consider the event of flipping a (possibly unfair) coin.
- $Y \in \{0, 1\}$  represents success and failure.
- Suppose we only flip the coin once,
  - We can express  $P(Y = 1) = p$  and  $P(Y = 0) = 1 - p$
- Bernoulli distribution

$$P(Y = y) = p^y(1 - p)^{1-y} \quad \text{for } y = 0, 1$$

- If we flip the coin  $n$  times,
- Binomial distribution

$$P(Y = y) = \binom{n}{y} p^y(1 - p)^{n-y} \quad \text{for } y = 0, 1, \dots, n$$

# Binomial distributions with different values of $n$ and $p$

If  $Y \sim \text{Binomial}(n, p)$ , then  $E(Y) = np$  and  $SD(Y) = \sqrt{np(1-p)}$ .

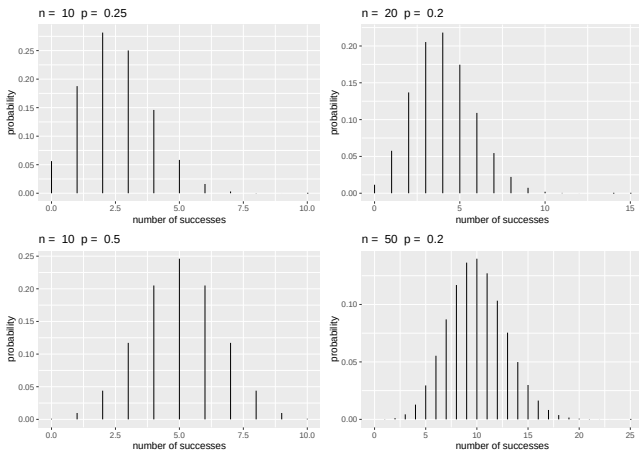


Figure 1: Binomial distributions with different values of  $n$  and  $p$ .

# How to generate in R?

All common distributions have four functions in R:

- Density  
`dbinom(x, size, prob)`
- Distribution function  
`pbinom(q, size, prob)`
- Quantile function  
`qbinom(p, size, prob)`
- Random generation  
`rbinom(n, size, prob)`

Not sure? Using `?` with any of the four functions, e.g. `?qbinom`

## Example of binomial distribution computing

**Question:** While taking a multiple choice test, a student encountered 10 problems where she ended up completely guessing, randomly selecting one of the four options. What is the chance that she got exactly 2 of the 10 correct?

## Example of binomial distribution computing

**Question:** While taking a multiple choice test, a student encountered 10 problems where she ended up completely guessing, randomly selecting one of the four options. What is the chance that she got exactly 2 of the 10 correct?

**Answer:** Knowing that the student randomly selected her answers, we assume she has a 25% chance of a correct response.

$$P(Y = 2) = \binom{10}{2} (.25)^2 (.75)^8 = 0.282$$



## Example of binomial distribution computing

**Question:** While taking a multiple choice test, a student encountered 10 problems where she ended up completely guessing, randomly selecting one of the four options. What is the chance that she got exactly 2 of the 10 correct?

**Answer:** Knowing that the student randomly selected her answers, we assume she has a 25% chance of a correct response.

$$P(Y = 2) = \binom{10}{2} (.25)^2 (.75)^8 = 0.282$$

**R computing:**

```
dbinom(2, size = 10, prob = .25)
```

```
## [1] 0.2815676
```

# Continuous random variable

A continuous random variable can take on an uncountably infinite number of values. Given a pdf  $f(y)$ ,

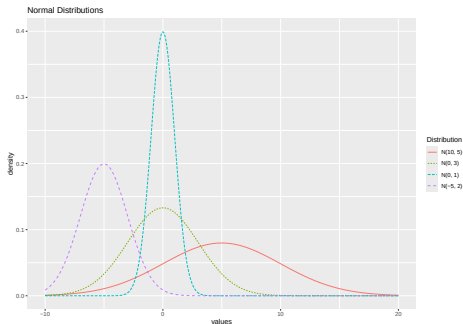
$$P(a \leq Y \leq b) = \int_a^b f(y)dy$$

Properties:

- $\int_{-\infty}^{\infty} f(y)dy = 1$ .
- For any value  $y$ ,  $P(Y = y) = \int_y^y f(y)dy = 0$ .  $P(y < Y) = P(y \leq Y)$ .

# Example of Continuous Distribution (Normal)

- The normal distribution is a very important distribution because:
  - A lot of things look normal
  - Analytically tractable
  - Central limit theorem
- $\frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right)$
- Characterized by mean,  $\mu$ , and variance,  $\sigma^2$ .



# How to Generate Samples from Normal Distribution

The following commands are for a normal random variable with mean  $\mu$  and variance  $\sigma^2$ , that is,  $X \sim N(\mu, \sigma^2)$ ,

- To calculate the probability density function at a value  $x$ ,  
`dnorm(x,mu,sigma)`
- To calculate the cumulative distribution function at a value  $x$ ,  
`pnorm(x,mu,sigma)`
- To generate a size  $m$  sample from the normal distribution,  
`rnorm(m,mu,sigma)`
- Note that the third argument is the **square root of the variance**, this is because the R function for normal distribution asks for the standard deviation, which is defined as the square root of the variance

# Some probability distributions in R

## Continuous

- Normal (`?rnorm`)
- Uniform (`?runif`)
- Beta (`?rbeta`)
- Chi-sq (`?rchisq`)
- Exponential (`?rexp`)
- t (`?rt`)
- F (`?rf`)
- Logistic (`?rlogis`)
- Lognormal (`?rlnorm`)

## Discrete

- Poisson (`?rpois`)
- Binomial (`?rbinom`)
- Geometric (`?rgeom`)
- Negative Binomial (`?rnbinom`)
- Multinomial (`?rmultinom`)

# Empirical vs. Theoretical CDF

In statistics, an empirical distribution function is the distribution function associated with the empirical measure of a sample.

- Theoretical CDF

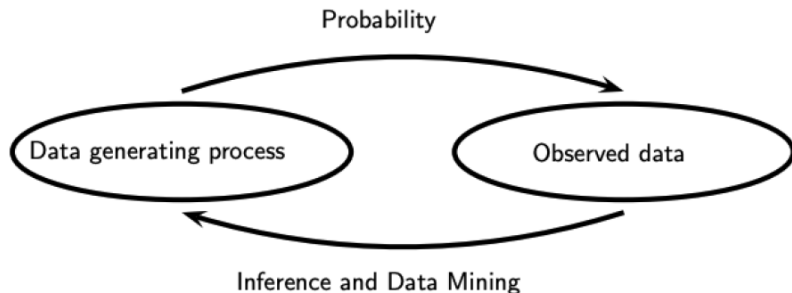
$$F_X(k) = \Pr(X \leq k)$$

- Empirical CDF

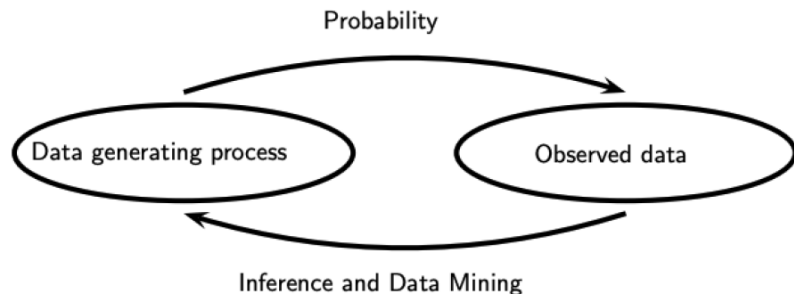
$$\hat{F}_n(k) = \frac{\text{number of elements in the sample} \leq k}{n} = \frac{1}{n} \sum_{i=1}^n I_{X_i \leq k}$$

where  $X_1, \dots, X_n$  make up some random sample from the underlying distribution.

# Probability and inference



# Probability and inference



- Probability: Given a data generating process, what are the properties of the outcomes?
- Statistical inference: Given the outcomes, what can we say about the process that generated the data?



# Parametric vs. Nonparametric models

- Statistical model  $\mathfrak{F}$ : a set of distributions (or densities or regression functions)

# Parametric vs. Nonparametric models

- Statistical model  $\mathfrak{F}$ : a set of distributions (or densities or regression functions)
- Parametric model: a set  $\mathfrak{F}$  that can be parameterized by a finite number of parameters

$$\mathfrak{F} = \{f(x; \theta) : \theta \in \Theta\}$$

where  $\theta$  is an unknown parameter (or vector of parameters) that can take values in the parameter space  $\Theta$ .

- e.g. Normal distribution, a 2-parameter model with density as  $f(x; \mu, \sigma)$

# Parametric vs. Nonparametric models

- Statistical model  $\mathfrak{F}$ : a set of distributions (or densities or regression functions)
- Parametric model: a set  $\mathfrak{F}$  that can be parameterized by a finite number of parameters

$$\mathfrak{F} = \{f(x; \theta) : \theta \in \Theta\}$$

where  $\theta$  is an unknown parameter (or vector of parameters) that can take values in the parameter space  $\Theta$ .

- e.g. Normal distribution, a 2-parameter model with density as  $f(x; \mu, \sigma)$
- Nonparametric model: a set  $\mathfrak{F}$  that cannot be parameterized by a finite number of parameters
  - e.g.  $\mathfrak{F}_{\text{ALL}} = \{\text{all CDF's}\}$  is nonparametric.

# Frequentist and Bayesian

- Frequentist: statistical methods with guaranteed frequency behavior
- Bayesian: statistical methods for using data to update beliefs

# Point estimation

- Providing a single “best guess” of some quantity of interest
- Notations
  - Parameter  $\theta$ : fixed, unknown quantity
  - Point estimator  $\hat{\theta}$ : depends on data, random variable

# Point estimation

- Providing a single “best guess” of some quantity of interest
- Notations
  - Parameter  $\theta$ : fixed, unknown quantity
  - Point estimator  $\hat{\theta}$ : depends on data, random variable

## Definition (Point estimator)

Let  $X_1, \dots, X_n$  be  $n$  IID data points from some distribution  $F$ . A point estimator  $\hat{\theta}_n$  of a parameter  $\theta$  is some function of  $X_1, \dots, X_n$  :

$$\hat{\theta}_n = g(X_1, \dots, X_n)$$

- What is a good point estimate?

- Definition:

$$\text{MSE} = \mathbb{E}_{\theta} \left( \hat{\theta}_n - \theta \right)^2$$

- No uniformly best estimator in terms of MSE
- It is NOT possible to have an estimator that is uniformly the best.

# Bias and Variance

- Bias

$$\text{bias}(\hat{\theta}_n) = \mathbb{E}_{\theta}(\hat{\theta}_n) - \theta$$

- Variance

$$\text{Var}(\hat{\theta}_n) = \mathbb{E}_{\theta}(\hat{\theta}_n - \mathbb{E}\theta)^2$$

- Theorem

$$MSE = \text{bias}^2 + \text{Var}$$



# Unbiasedness

- Definition

$$\text{bias}(\hat{\theta}_n) = \mathbb{E}_{\theta}(\hat{\theta}_n) - \theta = 0$$

- Unbiasedness is a small sample (finite sample) property
- An unbiased estimator may not exist
- An unbiased estimator is not necessarily a good estimator

# Consistency

- Definition

$$\hat{\theta}_n \xrightarrow{P} \theta$$

- It is possible to be unbiased but not consistent.
- It is possible to be consistent but not unbiased.

# Asypototic unbiasedness

- Definition

$$\text{bias}(\hat{\theta}_n) = \mathbb{E}_{\theta}(\hat{\theta}_n) - \theta \rightarrow 0, \text{ as } n \rightarrow \infty$$

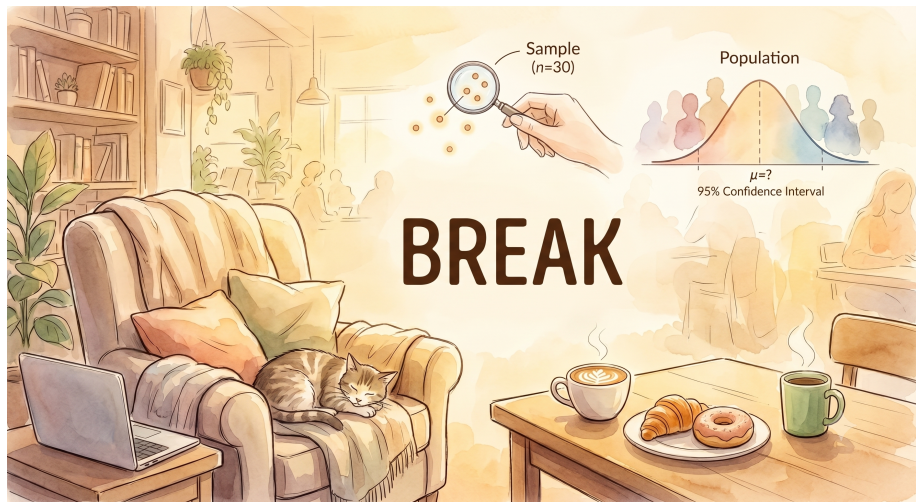
- It is possible to be asypototically unbiased but not consistent.
- It is possible to be consistent but NOT asymptotically unbiased.
- Sufficient conditions:  $MSE \rightarrow 0$ .

# Resources

This tutorial is based on

- Havard Biostatistics Summer Pre Course [\[link\]](#)
- “Beyond Multiple Linear Regression” by Paul Roback and Julie Legler [\[link\]](#)

# Break



# Outline

This module we will review

- Basics of parametric inference
- Methods for generating parametric estimators
- Maximum likelihood estimators
- Delta method
- Optimization method for finding MLE in R (Newton-Raphson, EM algorithm)

# Parametric inference

## Definition (Parametric models)

$$\mathfrak{F} = \{f(x; \theta) : \theta \in \Theta\}$$

where the  $\Theta \subset \mathbb{R}^k$  is the parameter space and  $\theta = (\theta_1, \dots, \theta_k)$  is the parameter.

## Goal of parametric inference

- estimate the parametric  $\theta$  (assume we known the form of the density).

# Parameter of interest and nuisance parameter

Often, we are interested in estimating some function  $T(\theta)$ .

For example, if  $X \sim N(\mu, \sigma^2)$ , then

- Parameters:  $\theta = (\mu, \sigma)$
- Parameter space:  $\Theta = \{(\mu, \sigma) : \mu \in \mathbb{R}, \sigma > 0\}$

If the goal is to estimate the  $\mu$  then

- Parameter of interest:  $T(\theta) = \mu$
- Nuisance parameter:  $\sigma$



# Methods for generating parametric estimators

- 1 Method of moments
- 2 Maximum likelihood

# Method of moments

## Definitions

- $\mathbb{E}(X^k)$  is the  $k^{th}$  (theoretical) moment of the distribution, for  $k = 1, 2, \dots$
- $M_k = \frac{1}{n} \sum_{i=1}^n X_i^k$  is the  $k^{th}$  sample moment, for  $k = 1, 2, \dots$

# Steps to find MoM

The basic idea behind this form of the method is to:

- Equate the first sample moment about the origin  $M_1 = \frac{1}{n} \sum_{i=1}^n X_i = \bar{X}$  to the first theoretical moment  $\mathbb{E}(X)$ .
- Equate the second sample moment about the origin  $M_2 = \frac{1}{n} \sum_{i=1}^n X_i^2$  to the second theoretical moment  $\mathbb{E}(X^2)$ .
- Continue equating sample moments about the origin until you have as many equations as you have parameters.
- Solve for the parameters

## Example of MoM (Bernoulli)

Let  $X_1, \dots, X_n$  be Bernoulli random variables with parameter  $p$ . What is the method of moments estimator of  $p$ ?

## Example of MoM (Bernoulli)

Let  $X_1, \dots, X_n$  be Bernoulli random variables with parameter  $p$ . What is the method of moments estimator of  $p$ ?

Only one parameter  $p$ , so we only need to equate the first moment.

$$\mathbb{E}(X_i) = p = M_1 = \frac{1}{n} \sum_{i=1}^n X_i = \bar{X}.$$

So,

$$\hat{p}_{MoM} = \bar{X}$$

## Example of MoM (Normal)

Let  $X_1, \dots, X_n$  be normal random variables with mean  $\mu$  and variance  $\sigma^2$ .  
What are the method of moments estimators of the mean  $\mu$  and variance  $\sigma^2$  ?

## Example of MoM (Normal)

Let  $X_1, \dots, X_n$  be normal random variables with mean  $\mu$  and variance  $\sigma^2$ . What are the method of moments estimators of the mean  $\mu$  and variance  $\sigma^2$  ?

Two parameters, so we need to equate the first and the second moment.

$$E(X_i) = \mu = M_1, E(X_i^2) = \sigma^2 + \mu^2 = M_2.$$

So,

$$\hat{\mu}_{MoM} = \bar{X}, \hat{\sigma}_{MoM}^2 = \frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}^2.$$

# Asymptotic properties

Under mild regularity conditions, MoM estimators are

- *Consistent*  $\rightarrow$  converge to the true value in probability as  $n \rightarrow \infty$ , i.e.

$$\lim_{n \rightarrow \infty} P(|\hat{\theta} - \theta| \leq \epsilon) = 1 \quad \forall \epsilon > 0$$

- *Asymptotically Normal*  $\rightarrow \sqrt{n}(\hat{\theta} - \theta) \sim N(0, \sigma^2)$  for large  $n$
- However, they are usually **NOT** *Asymptotically Efficient*



# Maximum likelihood

- Parametric model:  $f(x; \theta)$ ,  $X_1, \dots, X_n$  iid
- Likelihood function

$$\mathcal{L}_n(\theta) = \prod_{i=1}^n f(X_i; \theta)$$

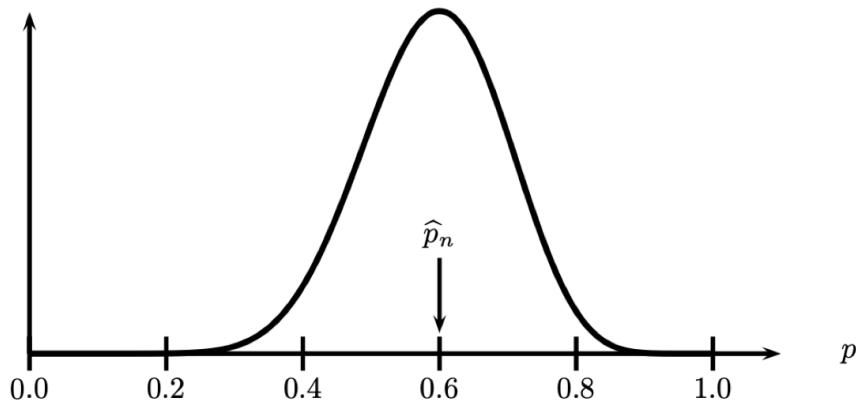
- The log-likelihood function

$$\ell_n(\theta) = \log \mathcal{L}_n(\theta) = \sum_{i=1}^n \log f(X_i; \theta)$$

- The maximum likelihood estimator (MLE)

$$\hat{\theta}_{MLE} = \arg \max_{\theta} \mathcal{L}(\theta)$$

## An example of MLE



Likelihood function for Bernoulli with  $n = 20$  and  $\sum_{i=1}^n X_i = 12$ . The MLE is  $\hat{p}_n = 12/20 = 0.6$ .

# Steps to find the MLE

- 1 Write out the likelihood

$$\mathcal{L}(\theta) = f(X_1, \dots, X_n; \theta)$$

- 2 Simplify the log likelihood

$$\ell(\theta) = \log \mathcal{L}(\theta)$$

- 3 Take the derivative of  $\ell(\theta)$  with respect to the parameter of interest,  $\theta$   
Set = 0
- 4 Solve for  $\theta$  (get  $\hat{\theta}_{MLE}$ )
- 5 Check that  $\hat{\theta}_{MLE}$  is a maximum ( $\frac{\partial^2}{\partial \theta^2} \ell(\theta) < 0$ )

## Exercise

Suppose we have an iid sample  $\{X_1, \dots, X_n\}$  with  $X_i \sim \text{Bernoulli}(p)$ . Find the MLE for  $p$ .

## Exercise

Suppose we have an iid sample  $\{X_1, \dots, X_n\}$  with  $X_i \sim \text{Bernoulli}(p)$ . Find the MLE for  $p$ .

1. The likelihood

$$\mathcal{L}_n(p) = \prod_{i=1}^n f(X_i; p) = \prod_{i=1}^n p^{X_i} (1-p)^{1-X_i} = p^S (1-p)^{n-S}$$

where  $S = \sum_i X_i$

## Exercise

Suppose we have an iid sample  $\{X_1, \dots, X_n\}$  with  $X_i \sim \text{Bernoulli}(p)$ . Find the MLE for  $p$ .

1. The likelihood

$$\mathcal{L}_n(p) = \prod_{i=1}^n f(X_i; p) = \prod_{i=1}^n p^{X_i} (1-p)^{1-X_i} = p^S (1-p)^{n-S}$$

where  $S = \sum_i X_i$

2. Log-likelihood

$$\ell_n(p) = S \log p + (n - S) \log(1 - p)$$

## Exercise

Suppose we have an iid sample  $\{X_1, \dots, X_n\}$  with  $X_i \sim \text{Bernoulli}(p)$ . Find the MLE for  $p$ .

1. The likelihood

$$\mathcal{L}_n(p) = \prod_{i=1}^n f(X_i; p) = \prod_{i=1}^n p^{X_i} (1-p)^{1-X_i} = p^S (1-p)^{n-S}$$

where  $S = \sum_i X_i$

2. Log-likelihood

$$\ell_n(p) = S \log p + (n - S) \log(1 - p)$$

3. MLE

$$\ell'_n(p) = 0$$

The MLE is  $\hat{p}_n = S/n$ .

# Asymptotics of MLE

Under mild regularity conditions, MLEs are

- *Consistent*  $\rightarrow$  converge to the true value in probability as  $n \rightarrow \infty$ , i.e.

$$\lim_{n \rightarrow \infty} P(|\hat{\theta} - \theta| \leq \epsilon) = 1 \quad \forall \epsilon > 0$$

- *Asymptotically normal*  $\rightarrow \sqrt{n}(\hat{\theta} - \theta) \sim N(0, \sigma^2)$  for large  $n$
- *Asymptotically efficient*
- *equivariant*  $\rightarrow$  if  $\hat{\theta}$  is the MLE for  $\theta$  then  $g(\hat{\theta})$  is the MLE for  $g(\theta)$



# Asymptotic Efficiency

## Cramér–Rao bound

The variance of any *unbiased* estimator  $\hat{\theta}$  of  $\theta$  is bounded by the reciprocal of the Fisher information  $I(\theta)$  :

$$\text{Var}(\hat{\theta}) \geq \frac{1}{I(\theta)},$$

where  $I(\theta) = n\mathbb{E} \left[ \left( \frac{\partial \ell}{\partial \theta} \right)^2 \right]$ .

Both MoM estimators are asymptotically unbiased, but MLE estimators achieves the CR lower bound.

Sometimes, there is no closed-form solution, so we need to use optimization methods to find the maximum of the log-likelihood.

- `optim()` find values of some parameters that **minimizes** some function.
- Newton-Raphson
- EM-algorithm (Not discussed here)

# Example using optim()

```
set.seed(42) # For reproducibility
sample_data <- rbinom(1000, size = 1, prob = 0.3) # Assuming success probability of 0.3

# Log-likelihood function for Bernoulli distribution
log_likelihood_bernoulli <- function(p, data) {
  n <- length(data)
  log_likelihood <- sum(data * log(p) + (1 - data) * log(1 - p))
  return(-log_likelihood) # Negative to be used with optimization functions (minimization)
}

# Initial parameter value for optimization (probability of success)
initial_param <- 0.8

# Find MLE using optim
result <- optim(
  par = initial_param, fn = log_likelihood_bernoulli,
  data = sample_data, method = "Brent", lower = 0, upper = 1
) # If bounds aren't provided, use method="BFGS"

# MLE estimate of p
mle_p <- result$par

# Print the result
cat("MLE of p:", mle_p, "\n")
```

```
## MLE of p: 0.293
```

# Newton-Raphson

Derivative of the log-likelihood around  $\theta^j$  :

$$0 = \ell'(\hat{\theta}) \approx \ell'(\theta^j) + (\hat{\theta} - \theta^j) \ell''(\theta^j)$$

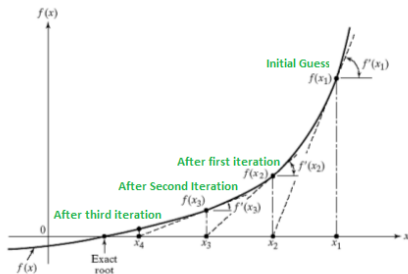
Solving for  $\hat{\theta}$  gives

$$\hat{\theta} \approx \theta^j - \frac{\ell'(\theta^j)}{\ell''(\theta^j)}.$$

This suggests the following iterative scheme:

$$\hat{\theta}^{j+1} = \theta^j - \frac{\ell'(\theta^j)}{\ell''(\theta^j)}$$

# Illustration



# NR algorithm in R

```
# First derivative of the log-likelihood function
log_likelihood_bernoulli_prime <- function(p, data) {
  n <- length(data)
  d_log_likelihood <- sum(data / p - (1 - data) / (1 - p))
  return(-d_log_likelihood) # Negative to be used with optimization functions (minimization)
}

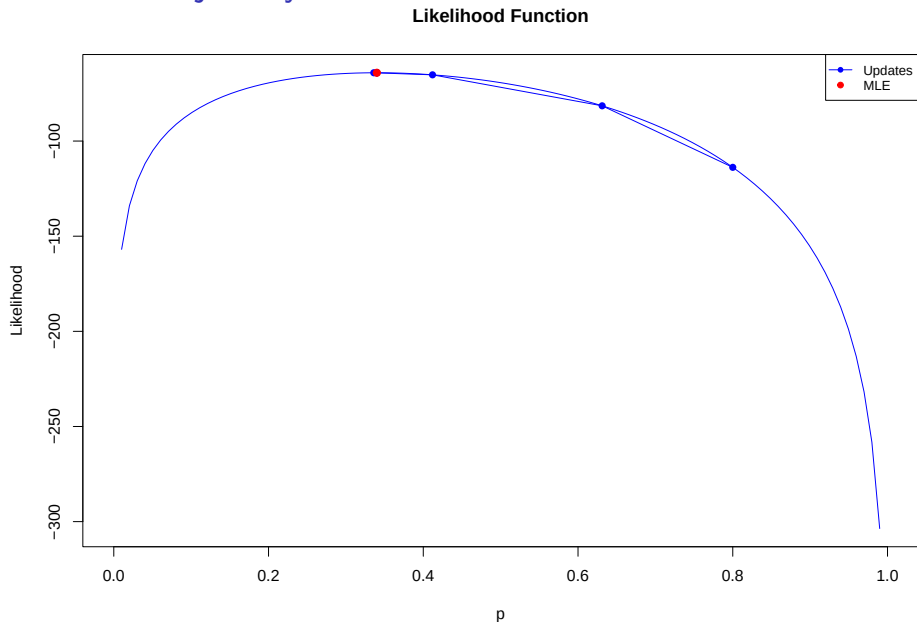
# Second derivative of the log-likelihood function
log_likelihood_bernoulli_double_prime <- function(p, data) {
  n <- length(data)
  dd_log_likelihood <- sum(-data / p^2 - (1 - data) / (1 - p)^2)
  return(-dd_log_likelihood) # Negative to be used with optimization functions (minimization)
}

# Initial parameter value for optimization (probability of success)
initial_param <- 0.8
# Newton-Raphson algorithm for optimization
tolerance <- 1e-8
max_iterations <- 1000
p <- initial_param
for (i in 1:max_iterations) {
  p_new <- p - log_likelihood_bernoulli_prime(p, sample_data) /
    log_likelihood_bernoulli_double_prime(p, sample_data)
  if (abs(p_new - p) < tolerance) {
    break
  }
  p <- p_new
}

# Print the result
cat("MLE of p:", p, "\n")
```

```
## MLE of p: 0.293
```

# Solution Trajectory



# Delta method

## Theorem (The Delta Method).

Suppose that

$$\frac{\sqrt{n}(Y_n - \mu)}{\sigma} \rightsquigarrow N(0, 1)$$

and that  $g$  is a differentiable function such that  $g'(\mu) \neq 0$ . Then

$$\frac{\sqrt{n}(g(Y_n) - g(\mu))}{|g'(\mu)|\sigma} \rightsquigarrow N(0, 1).$$

In other words,

$$Y_n \approx N\left(\mu, \frac{\sigma^2}{n}\right) \quad \text{implies that} \quad g(Y_n) \approx N\left(g(\mu), (g'(\mu))^2 \frac{\sigma^2}{n}\right).$$



## Exercise

Let  $X_1, \dots, X_n \sim \text{Bernoulli}(p)$  and let  $\psi = g(p) = \log(p/(1-p))$ . Find the MLE of  $\psi$  and its asymptotic distribution.

## Exercise

Let  $X_1, \dots, X_n \sim \text{Bernoulli}(p)$  and let  $\psi = g(p) = \log(p/(1-p))$ . Find the MLE of  $\psi$  and its asymptotic distribution.

The MLE of  $p$  was derived previously as follows

$$\hat{p}_{\text{MLE}} = \frac{1}{n} \sum_{i=1}^n X_i = \bar{X}$$

Now, for  $\psi = g(p) = \log\left(\frac{p}{1-p}\right)$ , the MLE of  $\psi$  is obtained by plugging in the MLE of  $p$ :

$$\hat{\psi}_{\text{MLE}} = g(\hat{p}_{\text{MLE}}) = \log\left(\frac{\hat{p}_{\text{MLE}}}{1 - \hat{p}_{\text{MLE}}}\right) = \log\left(\frac{\bar{X}}{1 - \bar{X}}\right)$$

## Exercise

By the Central Limit Theorem, as  $n \rightarrow \infty$ :

$$\sqrt{n}(\bar{X} - E[X_i]) \xrightarrow{D} N(0, \text{Var}(X_i))$$

Therefore,

$$\sqrt{n}(\hat{p}_{\text{MLE}} - p) \xrightarrow{D} N(0, p(1 - p))$$

To apply delta method, we need to find the derivative of  $g(p)$ :

$$g'(p) = \frac{1}{p(1 - p)}$$

and

$$[g'(p)]^2 \sigma^2 = \left( \frac{1}{p(1 - p)} \right)^2 \cdot p(1 - p) = \frac{1}{p^2(1 - p)^2} \cdot p(1 - p) = \frac{1}{p(1 - p)}$$

Therefore,

$$\sqrt{n}(\hat{\psi}_{\text{MLE}} - \psi) \xrightarrow{D} N\left(0, \frac{1}{p(1 - p)}\right)$$

# Resources

This tutorial is based on

- Harvard Biostatistics Summer Pre Course [\[link\]](#)
- “All of Statistics” by Larry A. Wasserman [\[link\]](#)

# Exercises

Available on course website.